AWANTIKA SRIVASTAVA

Data Scientist | Machine Learning, Data Science & NLP

+91-8920482037 |sawantika81@gmail.com| LINKEDIN |Github

PROFILE SUMMARY

Data Scientist with 2+ years of experience in Python, SQL, and machine learning for analyzing large-scale time-based and operational datasets. Skilled in fault data analysis, predictive modeling, and AWS analytics services including EC2, Athena, and SageMaker. Experienced in building end-to-end ML pipelines and generating data-driven insights to support business decision making.

CORE TECHNICAL SKILLS

- **Programming Languages:** Python, C++, SQL.
- **Statistics & Mathematics:** EDA, Statistical Modeling, Hypothesis Testing, A/B Testing, Probability, Linear Optimization, Trend Analysis.
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, Hyperparameter Tuning, Model Evaluation.
- **Deep Learning:** Neural Networks, CNN, RNN, LSTM, Transformers (BERT), Transfer Learning, Model Fine-tuning.
- **Computer Vision**: Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Video Analytics, OpenCV, Paddle OCR.
- **NLP**: Text preprocessing, Tokenization, Transformers, Sentiment Analysis, Topic Modeling (LSA, LDA), Transformer-based models.
- **Time-Series & Fault Analysis:** Time-based Data Analysis, Anomaly Detection, Fault Pattern Identification, ARIMA, SARIMA
- **Cloud Platforms:** AWS EC2, AWS Athena, AWS SageMaker, AWS Glue (basic familiarity)
- **Tools & Visualization:** Jupyter Notebook, Colab, Git, GitHub, Linux, Matplotlib, Seaborn

EXPERIENCE

Data Scientist | PPS International Pvt. Ltd. **January 2024-Present**

- Performed **data analysis on large structured datasets using Python and SQL** to identify patterns and operational insights.
- Built **machine learning models for predictive analytics**, including regression and classification algorithms.
- Conducted **time-based data analysis and anomaly detection** to identify abnormal patterns in system data.
- Developed scripts using **Python (Pandas, NumPy)** for data cleaning, transformation, and statistical analysis.
- Designed and implemented **end-to-end machine learning pipelines**, covering data ingestion, preprocessing, model training, evaluation, and deployment workflows.
- Automated **data processing and reporting workflows using Python scripts and SQL queries**, improving data analysis efficiency and reducing manual processing time.
- Collaborated with cross-functional teams to understand requirements and deliver **data-driven insights**.
- Built **data visualization dashboards and reports** to communicate analytical findings to stakeholders.
- Performed time-based and fault data analysis to detect anomalies and system performance issues.
- Worked with AWS analytics services such as Athena and SageMaker for scalable data processing.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- Designed and deployed a **real-time ML systems** for unsafe behavior detection using **CNN-based object detection** models.
- Implemented **end-to-end ML pipelines** from data ingestion to production inference.
- **Deployed** optimized models using **TensorFlow Lite** for continuous, **low-latency inference.**
- Developed a **Flask-based** web interface to display detections and recorded video clips in **real time**.
- **Deployed** the complete solution on **edge devices**, enabling reliable on-device inference and **real-time event recordings**.

Amazon Stock Price prediction | Applied ML Project

- **Built batch-oriented ML workflows** for time-series forecasting using **LSTM** models.
- Implemented data **preprocessing**, feature engineering, and model **evaluation pipelines** on **large** historical datasets.
- Designed **sliding-window** based **sequence generation** and trained LSTM model 5-year stock price data.
- Evaluated model performance using appropriate regression **metrics** and **trend-based analysis** for short-term forecasting.

YouTube Comments Sentiment Analyzer | link - https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Deployed transformer-based **NLP models** as **production-ready** services with **REST APIs**.
- **Fine-tuned** and served a **DistilBERT**-based **sentiment** classification model for large-scale text inference.
- **Built** and deployed an interactive **streamlit web application** to perform real-time **sentiment analysis** on YouTube comments.
- **Processed** high-volume text **data** with sub-second inference **latency** for real-time sentiment analysis.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**

Bachelor of Technology (Electrical and electronics engineering)